

Pay, Oscillate, or Learn?

Three Agent Decision Rules under Congestion Pricing

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Agent-Based Modelling

Q-Learning

El Farol

Auckland CBD

ToU Pricing

Why Does the Behavioural Assumption Matter?



Auckland's Context

Congestion costs NZ\$2.6B/yr. Time-of-Use (ToU) charging legislation passed Nov 2025. CBD cordon launch expected alongside City Rail Link.



The Gap

Most models ask 'how much will traffic fall?' — but behavioural assumptions drive whether congestion shifts in time, in space, or not at all.



Our Approach

ABM with 500 heterogeneous agents, Auckland CBD road network. Three decision rules, same pricing environment, same V/C measurement.

Three Agent Decision Rules: A Rationality Spectrum

← Static Rationality

01

Exponential Decay

Static Rationality

$$P(\text{entry}) = e^{-(\alpha \cdot \text{fee} / \text{VoT})}$$

- Entry probability declines with fee/VoT ratio
- No memory, no adaptation
- Standard equilibrium assumption
- Result: smooth deterrence, diminishing returns

02

El Farol Bar Model

Bounded Rationality (No Learning)

Enter if best predictor: $E[\text{congestion}] < \theta$

- Portfolio of predictors based on recent history
- Coordination paradox: over-enter then over-avoid
- Oscillatory V/C, fee signal largely ignored
- Result: V/C edges upward under ToU (counter-intuitive)

Bounded Rationality with Learning →

03

Q-Learning

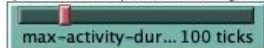
Bounded Rationality with Learning

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)]$$

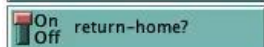
- Learns from own departure time experience
- Reward penalises fee paid + congestion encountered
- Converges within ~10 simulated days
- Result: peak V/C $0.65 \rightarrow 0.39$; peak entry $77.7\% \rightarrow 41.7\%$



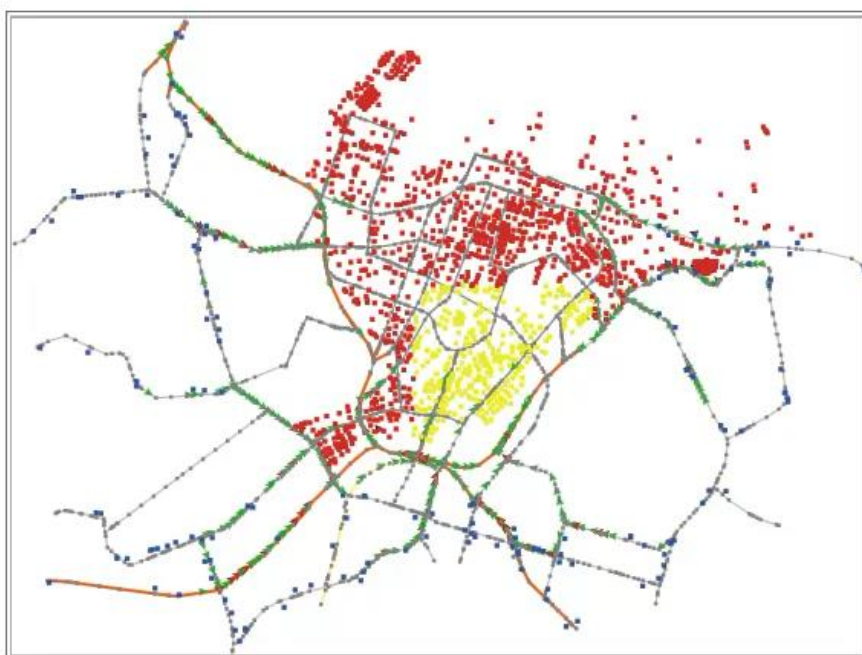
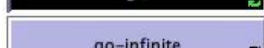
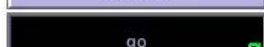
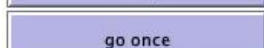
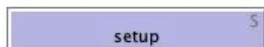
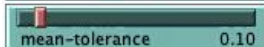
Maximum number of ticks a vehicle spends at an activity (i.e., at a building)



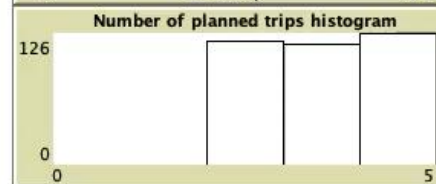
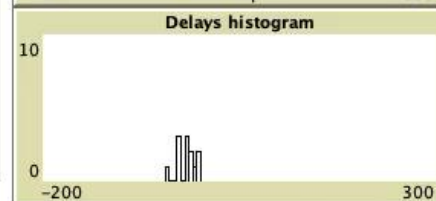
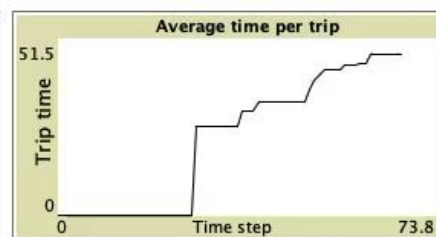
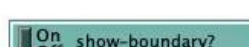
Maximum number of ticks that the vehicle moves earlier than it should to reach its next building destination



Minimum delay threshold at which an activity modification is requested is only reached if w



Control the seed value (Note that go-infinite increments the current-seed)



What Happens When You Charge Drivers?

Same road. Same price. Three very different outcomes — depending on how drivers think.



Baseline Driver

"I check the price and decide."

Fewer drivers enter at peak, but the effect is small. Once price-sensitive drivers have already left, adding a higher charge changes little.

Peak V/C (inner cordon)

No charge 0.38

ToU 0.36

→ Mild reduction



Herd Driver

"Everyone avoided it yesterday, so I'll go today."

Drivers react to what others did — not the price. This creates a loop: roads empty, then flood, then empty again. Congestion actually gets slightly worse.

Peak V/C (inner cordon)

No charge 0.58

ToU 0.61

↑ Backfires



Learning Driver

"I'll try leaving earlier — it worked last week."

Drivers gradually learn from experience. Within about 10 days, peak-hour traffic drops by nearly half. Crucially, congestion does not simply shift to nearby roads.

Peak V/C (inner cordon)

No charge 0.65

ToU 0.39

↓ Clear improvement

Key insight: the right behavioural assumption changes not just how much congestion falls — but whether the policy works at all.

Conclusions & Next Steps

Key Findings

- 1 Behavioural assumption shapes both magnitude and temporal redistribution of congestion — not just the quantity
- 2 Q-learning produces the most policy-relevant outcome: sustained peak reduction without spatial displacement
- 3 El Farol reveals the danger of congestion policies when agents ignore fee signals in favour of herd prediction
- 4 ABM places competing theories side-by-side within a common environment — revealing structural differences invisible to aggregate models

Next Steps

Model Extension

Mode choice, household VoT calibration from NZ travel survey data

LLM-ABM Hybrid

3-layer cognitive architecture: reactive (classical), deliberate (lightweight LLM), reflective (daily/weekly reasoning)

Sensitivity Analysis

Key behavioural parameters: α in decay, memory window in El Farol, learning rate and ϵ in Q-learning

Scale Up

Expand from 500 to ~70,000 agents; dynamic OD matrix replacing fixed assumption